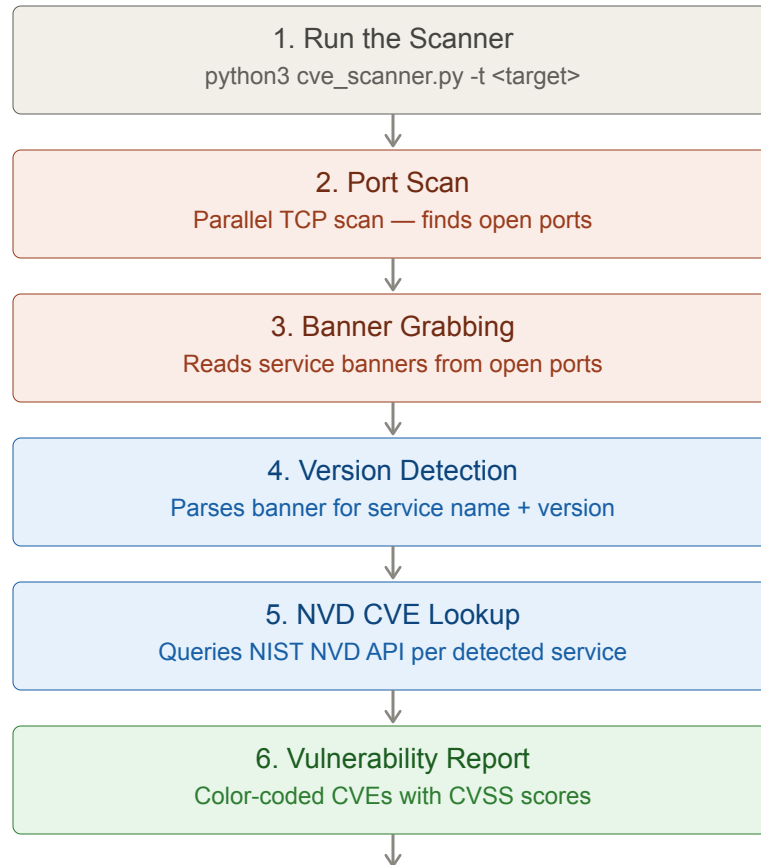


CVE Scanner — cheat sheet

Scans a target, detects service versions via banner grabbing, and queries NIST NVD for known CVEs.



Step 1 — Install & Run

```
pip install -r requirements.txt

# Interactive mode
python3 cve_scanner.py

# Direct scan
python3 cve_scanner.py -t 192.168.1.1

# Custom ports
python3 cve_scanner.py -t 192.168.1.1 -p 22,80,443,8080

# Top-1000 scan
python3 cve_scanner.py -t 192.168.1.1 --top1000
```

Step 2 — Port Scan

Scans common TCP ports in parallel using a thread pool. Default set covers 22 high-value ports (SSH, HTTP, HTTPS, SMB, RDP, MySQL, etc.). Use `-p` for custom ports or `--top1000` for full coverage.

Step 3 & 4 — Banner Grabbing & Version Detection

Connects to each open port and reads what the service sends back. Parses the banner with regex to extract:

Service	What it looks for
OpenSSH	SSH-2.0-OpenSSH_8.9p1 → OpenSSH 8.9p1
Apache	Server: Apache/2.4.51 → Apache 2.4.51
nginx	Server: nginx/1.18.0 → nginx 1.18.0
MySQL	5.7.38-MySQL Community → MySQL 5.7.38
ProFTPD	220 ProFTPD 1.3.5e → ProFTPD 1.3.5e

Step 5 — NVD CVE Lookup

Queries the free NIST NVD API for each detected service + version. Returns up to 5 CVEs per service with:

Field	Description
CVE ID	e.g. CVE-2021-41617
CVSS Score	0.0 – 10.0
Severity	CRITICAL / HIGH / MEDIUM / LOW
Description	Short summary of the vulnerability
Link	Direct URL to the NVD page

Step 6 — Reading the Report

```
[CRITICAL] CVE-2021-44228  Score: 10.0
  Apache Log4j2 – Remote Code Execution (Log4Shell)
  https://nvd.nist.gov/vuln/detail/CVE-2021-44228
```

```
[HIGH]      CVE-2022-22963  Score: 9.8
  Spring Framework RCE via routing expressions
```

Feed CVE IDs directly into exploit-suggester to find matching PoCs.

Key idea: this tool gives you a fast CVE map without touching a vulnerability scanner like Nessus. Feed CVE IDs into exploit-suggester for the next step.